- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

3 Forces $\mathbf{F}_1 = (2\mathbf{i} + 9\mathbf{j})$ N and $\mathbf{F}_2 = (-\mathbf{i} + \mathbf{j})$ N act on a particle. A third force $\mathbf{F}_3$ acts so that the particle is in equilibrium under the action of the three forces.

Find the force $\mathbf{F}_3$. **[2]**